

Solarise E-Commerce - Project Completion Report

Date: April 28, 2026 | Developer: Paras Ghimire | Status: DEPLOYED

1. Project Overview

- Full-stack solar energy e-commerce platform (Solarise)
- Django REST Framework backend + Next.js 14 frontend
- Deployed on DigitalOcean Droplet (138.197.11.80) via Docker
- GitHub repo: https://github.com/developerparasghimire/frontend_eco

2. Backend - Django REST Framework

- Custom User model with JWT authentication (SimpleJWT)
- Password reset via email token flow
- Products app: Category, Product with specs, pricing, inventory, images
- Orders app: Order lifecycle, order items, shipping address, signals
- Coupons app: Discount code validation with expiry and usage limits
- Reviews app: Star-rating reviews tied to verified purchases
- Wishlists app: Per-user product wishlist
- Contacts app: Contact form submission endpoint
- PayPal payment integration (orders/services/paypal.py)
- drf-spectacular: Auto-generated OpenAPI docs
- WhiteNoise for static file serving
- Gunicorn WSGI server with 2 workers (optimised for 512MB RAM)
- SQLite database persisted via Docker named volume
- Full test suite: 10 test files covering all apps + security tests

3. Frontend - Next.js 14 (App Router)

- Standalone output mode for Docker multi-stage build
- Tailwind CSS for responsive styling
- Auth: Login, Register, Forgot Password, Reset Password pages
- Shop: Products listing with filters, Product detail with gallery
- Cart: Add to cart, quantity management, coupon code input
- Checkout: Address picker, PayPal payment integration
- Dashboard: Orders history, Addresses management, Change password
- Wishlist page with add/remove functionality
- Home page, Services, Projects, News, Team pages
- Zustand store for auth state management
- Custom hooks: useProducts, useOrders, useCart, useWishlist, useReviews
- Security headers: X-Frame-Options, CSP, Referrer-Policy
- Recently viewed products (localStorage)

4. DevOps & Deployment

- Two Docker images: solar-backend (Python 3.12 slim) + solar-frontend (Node 20 Alpine multi-stage)
- Single docker-compose.yml orchestrates backend + frontend + nginx
- Nginx reverse proxy: / -> frontend:3000, /api/ -> backend:8000

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- Docker named volumes: sqlite_data, static_data, media_data
- Health checks on all services before nginx starts
- Environment via .env file (secrets never committed to git)
- .gitignore: excludes .venv, node_modules, db.sqlite3, .next, .env
- GitHub push: all code at https://github.com/developerparasghimire/frontend_eco
- DigitalOcean Droplet \$12/month (1 vCPU, 512MB RAM, NYC3)

5. Live URLs

Frontend:	http://138.197.11.80
Backend API:	http://138.197.11.80/api/
Admin Panel:	http://138.197.11.80/api/admin/
GitHub:	https://github.com/developerparasghimire/frontend_eco

6. Seed Data Added

- 4 categories: Solar Panels, Inverters, Batteries, Accessories
- 3 Solar Panels: SunPower 540W, Canadian Solar 400W, Jinko 450W Bifacial
- 3 Inverters: SolarEdge 5kW Hybrid, Fronius 8kW Grid, Growatt 3kW Off-Grid
- 2 Batteries: Tesla Powerwall 13.5kWh, Pylontech 5kWh LiFePO4
- 2 Accessories: MC4 Connector Kit, Mounting Rail 3m